

Deep Generative Models

Lecture 11

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AI Masters

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Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}_\theta(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

Theorem (Continuity Equation)

If $\mathbf{f}$ is uniformly Lipschitz continuous in $\mathbf{x}$ and continuous in t , then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left(\frac{\partial \mathbf{f}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

Solution of the Continuity Equation

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \text{tr} \left(\frac{\partial \mathbf{f}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt.$$

- ▶ This solution gives us the density along the trajectory (not the total probability path).
- ▶ However, it's difficult to efficiently estimate **the last term**.

Recap of Previous Lecture

SDE Basics

Let's define a stochastic process $\mathbf{x}(t)$ with initial condition $\mathbf{x}(0) \sim p_0(\mathbf{x})$:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w},$$

where $\mathbf{w}(t)$ is the standard Wiener process (Brownian motion):

$$\mathbf{w}(t) - \mathbf{w}(s) \sim \mathcal{N}(0, (t-s)\mathbf{I}), \quad d\mathbf{w} = \epsilon \cdot \sqrt{dt}, \text{ where } \epsilon \sim \mathcal{N}(0, \mathbf{I}).$$

Discretization of SDE (Euler Method) - SDEsolve

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \epsilon \cdot \sqrt{dt}$$

- ▶ At each time t , we have the density $p_t(\mathbf{x}) = p(\mathbf{x}, t)$.
- ▶ $p : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}_+$ is a **probability path** between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.

Recap of Previous Lecture

Theorem (Kolmogorov-Fokker-Planck)

The evolution of the distribution $p_t(\mathbf{x})$ is given by:

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

Langevin SDE (Special Case)

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) dt + \mathbf{1} \cdot d\mathbf{w}$$

The density $p_{\theta}(\mathbf{x})$ is a **stationary** distribution for the SDE.

Langevin Dynamics

Samples from the following dynamics will come from $p_{\theta}(\mathbf{x})$ under mild regularity conditions for a small enough η :

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\eta}{2} \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \sqrt{\eta} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}).$$

Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

Variance Exploding SDE (NCSN)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}, \quad \mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

The variance grows since $\sigma(t)$ is a monotonically increasing function.

Variance Preserving SDE (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

The variance is preserved if $\mathbf{x}(0)$ has unit variance.

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

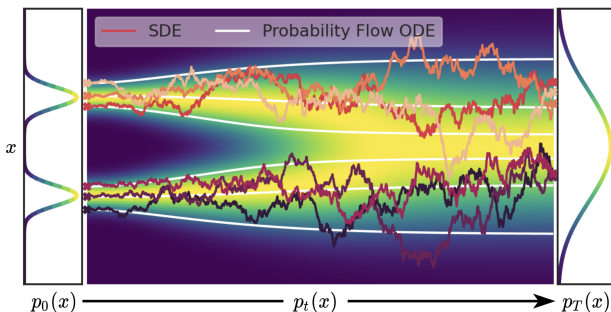
Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad (\text{SDE with the probability path } p_t(\mathbf{x}))$$

Probability Flow ODE

There exists an ODE with the identical probability path $p_t(\mathbf{x})$ of the form:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$



Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

Outline

1. Reverse SDE
2. Score-Based Generative Models Through SDEs
3. Flow Matching
4. Conditional Flow Matching

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1. Reverse SDE
2. Score-Based Generative Models Through SDEs
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Reverse SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}, t)dt$$

Here dt can be > 0 or < 0 .

Reverse SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}, t)dt$$

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Reverse ODE

Let $\tau = 1 - t$ ($d\tau = -dt$).

$$d\mathbf{x} = -\mathbf{f}(\mathbf{x}, 1 - \tau)d\tau$$

Reverse SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}, t)dt$$

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Reverse ODE

Let $\tau = 1 - t$ ($d\tau = -dt$).

$$d\mathbf{x} = -\mathbf{f}(\mathbf{x}, 1 - \tau)d\tau$$

- ▶ How do we reverse the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$?
- ▶ The Wiener process introduces randomness that must be reversed.

Reverse SDE

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- ▶ The Wiener process introduces randomness that must be reversed.

Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$, given by:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + g(t)d\bar{\mathbf{w}}$$

where $dt < 0$ and $\bar{\mathbf{w}}$ is a reverse-time Wiener process.

Reverse SDE

Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$, given by:

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where $dt < 0$ and $\bar{\mathbf{w}}(t)$ is a reverse-time Wiener process.

Note: Again, the score function appears: $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$.

Proof Sketch

- ▶ Convert the initial SDE to a probability flow ODE.
- ▶ Reverse the probability flow ODE.
- ▶ Convert the reversed probability flow ODE back to an SDE.

Proof of Reverse SDE

- Convert the initial SDE to a probability flow ODE:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

Proof of Reverse SDE

- Convert the initial SDE to a probability flow ODE:

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- Reverse the probability flow ODE:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1 - \tau) + \frac{1}{2}g^2(1 - \tau)\frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau$$

Proof of Reverse SDE

- Convert the initial SDE to a probability flow ODE:

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- Convert the reversed probability flow ODE back to an SDE:

$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1 - \tau) + \frac{1}{2}g^2(1 - \tau)\frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau$$

$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1 - \tau) + g^2(1 - \tau)\frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau + g(1 - \tau)d\mathbf{w}$$

Here $\mathbf{w}(\tau)$ is a Wiener process in forward τ -time.

Reverse SDE

Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$, given by:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + g(t) d\bar{\mathbf{w}}$$

where $dt < 0$ and $\bar{\mathbf{w}}(t)$ is a reverse-time Wiener process.

Proof (Continued)

$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1 - \tau) + g^2(1 - \tau) \frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau + g(1 - \tau) d\mathbf{w}$$

Reverse SDE

Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$, given by:

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Proof (Continued)

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$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + g(t) d\bar{\mathbf{w}}$$

Here $d\tau > 0$, $dt < 0$, and $\bar{\mathbf{w}}$ is reverse-time Wiener process.

Reverse SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad - \text{SDE}$$

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt \quad - \text{Probability Flow ODE}$$

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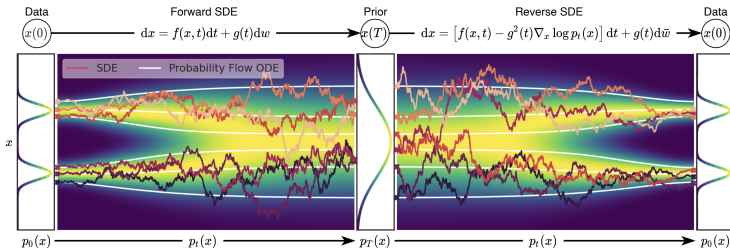
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- ▶ This framework allows us to transform one distribution into another via an SDE with a prescribed probability path $p_t(\mathbf{x})$.
- ▶ We can invert this process using the score function.



Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

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Variance Exploding SDE

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$\sigma(t)$ is a monotonically increasing function.

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$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}, \quad \mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

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$$d\mathbf{x} = \left(-\frac{1}{2} \frac{d[\sigma^2(t)]}{dt} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt \quad (\text{probability flow ODE})$$

$$d\mathbf{x} = \left(-\frac{d[\sigma^2(t)]}{dt} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + \sqrt{\frac{d[\sigma^2(t)]}{dt}} d\bar{\mathbf{w}} \quad (\text{reverse SDE})$$

Diffusion SDE

Variance Preserving SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

Variance is preserved as long as $\mathbf{x}(0)$ has unit variance.

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Outline

1. Reverse SDE
2. Score-Based Generative Models Through SDEs
3. Flow Matching
4. Conditional Flow Matching

Score-Based Generative Models Through SDEs

Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

Is it possible to train score-based diffusion models in continuous time?

Score-Based Generative Models Through SDEs

Discrete-Time Objective

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Is it possible to train score-based diffusion models in continuous time?

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0, 1]} \mathbb{E}_{q(\mathbf{x}(t) | \mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t) | \mathbf{x}(0)) \right\|_2^2$$

Score-Based Generative Models Through SDEs

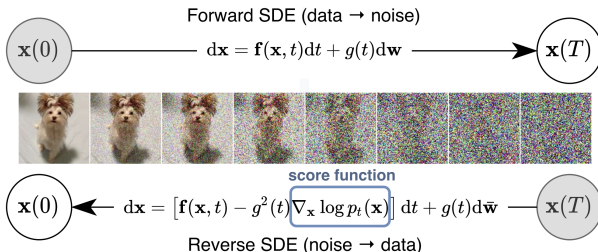
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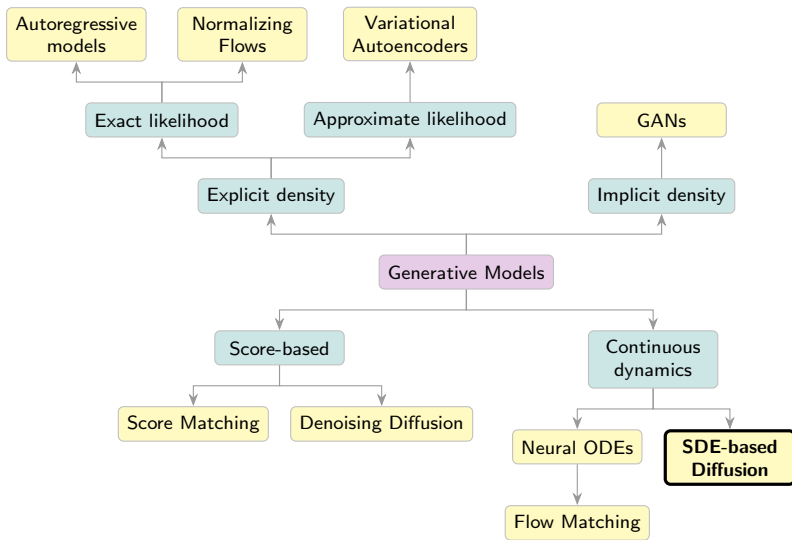
Continuous-Time Objective

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Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

Generative Models Taxonomy



Score-Based Generative Models Through SDEs

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

Score-Based Generative Models Through SDEs

Continuous-Time Objective

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$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(t, \mathbf{x}(0)), \boldsymbol{\sigma}^2(t, \mathbf{x}(0)) \cdot \mathbf{I}\right)$$

$$\nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) = -\frac{1}{\boldsymbol{\sigma}} \odot (\mathbf{x}(t) - \boldsymbol{\mu})$$

Note: Normality holds for $\mathbf{f}(\mathbf{x}, t)$ affine in $\mathbf{x}$ ($\mathbf{f}(\mathbf{x}, t) = f(t)\mathbf{x}$).

Score-Based Generative Models Through SDEs

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

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$$\nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) = -\frac{1}{\boldsymbol{\sigma}} \odot (\mathbf{x}(t) - \boldsymbol{\mu})$$

Note: Normality holds for $\mathbf{f}(\mathbf{x}, t)$ affine in $\mathbf{x}$ ($\mathbf{f}(\mathbf{x}, t) = f(t)\mathbf{x}$).

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w} \quad (\text{Variance Exploding SDE})$$

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w} \quad (\text{Variance Preserving SDE})$$

Is it possible to explicitly derive $\boldsymbol{\mu}(t, \mathbf{x}(0))$ and $\boldsymbol{\Sigma}(t, \mathbf{x}(0))$ for VE-SDE and VP-SDE?

Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(t, \mathbf{x}(0)), \boldsymbol{\Sigma}(t, \mathbf{x}(0))\right)$$

Theorem

The moments of the SDE $d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}$ satisfy:

$$\frac{d\boldsymbol{\mu}(t, \mathbf{x}(0))}{dt} = f(t)\boldsymbol{\mu}(t, \mathbf{x}(0))$$

$$\frac{d\boldsymbol{\Sigma}(t, \mathbf{x}(0))}{dt} = 2f(t)\boldsymbol{\Sigma}(t, \mathbf{x}(0)) + g^2(t) \cdot \mathbf{I}$$

Score-Based Generative Models Through SDEs

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Proof

$$\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] = \mathbb{E}[f(t)\mathbf{x}dt|\mathbf{x}(0)] + \mathbb{E}[g(t)d\mathbf{w}|\mathbf{x}(0)]$$

Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(t, \mathbf{x}(0)), \boldsymbol{\Sigma}(t, \mathbf{x}(0))\right)$$

Theorem

The moments of the SDE $d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}$ satisfy:

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$$\frac{d\boldsymbol{\Sigma}(t, \mathbf{x}(0))}{dt} = 2f(t)\boldsymbol{\Sigma}(t, \mathbf{x}(0)) + g^2(t) \cdot \mathbf{I}$$

Proof

$$\begin{aligned}\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] &= \mathbb{E}[f(t)\mathbf{x}dt|\mathbf{x}(0)] + \mathbb{E}[g(t)d\mathbf{w}|\mathbf{x}(0)] \\ &= f(t)\mathbb{E}[\mathbf{x}|\mathbf{x}(0)]dt + g(t)\mathbb{E}[d\mathbf{w}|\mathbf{x}(0)]\end{aligned}$$

Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(t, \mathbf{x}(0)), \boldsymbol{\Sigma}(t, \mathbf{x}(0))\right)$$

Theorem

The moments of the SDE $d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}$ satisfy:

$$\frac{d\boldsymbol{\mu}(t, \mathbf{x}(0))}{dt} = f(t)\boldsymbol{\mu}(t, \mathbf{x}(0))$$

$$\frac{d\boldsymbol{\Sigma}(t, \mathbf{x}(0))}{dt} = 2f(t)\boldsymbol{\Sigma}(t, \mathbf{x}(0)) + g^2(t) \cdot \mathbf{I}$$

Proof

$$\begin{aligned}\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] &= \mathbb{E}[f(t)\mathbf{x}dt|\mathbf{x}(0)] + \mathbb{E}[g(t)d\mathbf{w}|\mathbf{x}(0)] \\ &= f(t)\mathbb{E}[\mathbf{x}|\mathbf{x}(0)]dt + g(t)\mathbb{E}[d\mathbf{w}|\mathbf{x}(0)] \\ &= f(t)\boldsymbol{\mu}(t, \mathbf{x}(0))dt\end{aligned}$$

Score-Based Generative Models Through SDEs

Theorem

$$\frac{d\mu(t, \mathbf{x}(0))}{dt} = f(t)\mu(t, \mathbf{x}(0))$$

Score-Based Generative Models Through SDEs

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Proof (Continued)

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Score-Based Generative Models Through SDEs

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Examples

$$\text{NCSN: } f(t) = 0 \quad \Rightarrow \quad \mu = \mathbf{x}(0)$$

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$$\frac{d\boldsymbol{\mu}(t, \mathbf{x}(0))}{dt} = f(t)\boldsymbol{\mu}(t, \mathbf{x}(0))$$

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Examples

$$\textbf{NCSN: } f(t) = 0 \quad \Rightarrow \quad \boldsymbol{\mu} = \mathbf{x}(0)$$

$$\textbf{DDPM: } f(t) = -\frac{1}{2}\beta(t) \quad \Rightarrow \quad \frac{d\boldsymbol{\mu}}{dt} = -\frac{1}{2}\beta(t)\boldsymbol{\mu}$$

Score-Based Generative Models Through SDEs

Theorem

$$\frac{d\mu(t, \mathbf{x}(0))}{dt} = f(t)\mu(t, \mathbf{x}(0))$$

Proof (Continued)

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Examples

$$\textbf{NCSN: } f(t) = 0 \quad \Rightarrow \quad \mu = \mathbf{x}(0)$$

$$\textbf{DDPM: } f(t) = -\frac{1}{2}\beta(t) \quad \Rightarrow \quad \frac{d\mu}{dt} = -\frac{1}{2}\beta(t)\mu$$

$$\mu = \mathbf{x}(0) \exp\left(-\frac{1}{2} \int_0^t \beta(s) ds\right)$$

Score-Based Generative Models Through SDEs

Training

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(t, \mathbf{x}(0)), \boldsymbol{\Sigma}(t, \mathbf{x}(0))\right)$$

Score-Based Generative Models Through SDEs

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NCSN

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0), [\sigma^2(t) - \sigma^2(0)] \cdot \mathbf{I}\right)$$

Score-Based Generative Models Through SDEs

Training

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(t, \mathbf{x}(0)), \boldsymbol{\Sigma}(t, \mathbf{x}(0))\right)$$

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DDPM

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0)e^{-\frac{1}{2} \int_0^t \beta(s) ds}, \left(1 - e^{-\int_0^t \beta(s) ds}\right) \cdot \mathbf{I}\right)$$

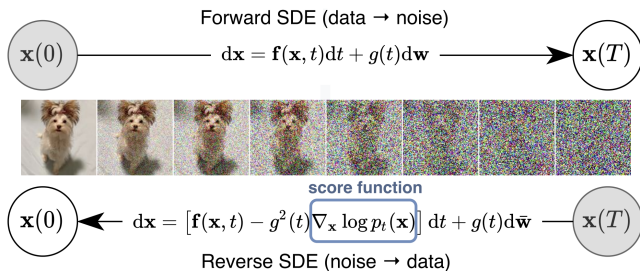
Here we omit the derivations of the variance.

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Score-Based Generative Models Through SDEs

Sampling

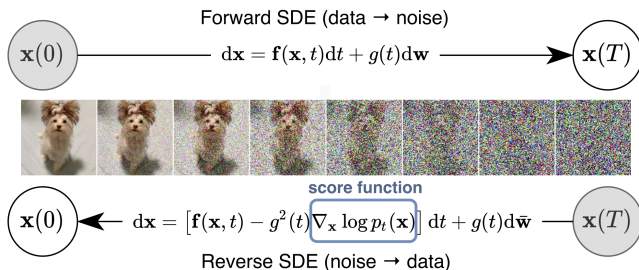
Solve the reverse SDE using numerical solvers (SDEsolve).



Score-Based Generative Models Through SDEs

Sampling

Solve the reverse SDE using numerical solvers (SDEsolve).



- ▶ Discretizing the reverse SDE provides ancestral sampling.
- ▶ Discretizing the probability flow ODE yields deterministic sampling.

Outline

1. Reverse SDE
2. Score-Based Generative Models Through SDEs
3. Flow Matching
4. Conditional Flow Matching

Continuous-Time Normalizing Flows

Let's return to ODE dynamics $\mathbf{x}_t = \mathbf{x}(t)$ in the interval $t \in [0, 1]$:

- ▶ $\mathbf{x}_0 \sim p_0(\mathbf{x}) = p(\mathbf{x})$, $\mathbf{x}_1 \sim p_1(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$;
- ▶ $p(\mathbf{x})$ is a base distribution (e.g., $\mathcal{N}(0, \mathbf{I})$), and $p_{\text{data}}(\mathbf{x})$ is the true data distribution.

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

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$$\frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left(\frac{\partial \mathbf{f}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

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- ▶ It's hard to solve the continuity equation directly due to the trace term.
- ▶ There's a method (the adjoint method) that solves this equation directly, but it's unstable and unscalable.

Continuous-Time Normalizing Flows

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{f}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

- ▶ Knowing the vector field $\mathbf{f}(\mathbf{x}, t)$, the KFP (or continuity) equation allows us to compute the density $p_t(\mathbf{x})$.
- ▶ Flow matching provides an alternative approach to Neural ODEs.

Continuous-Time Normalizing Flows

KFP Theorem (Continuity Equation)

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Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{f}(\mathbf{x}, t) - \mathbf{f}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Continuous-Time Normalizing Flows

KFP Theorem (Continuity Equation)

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Flow Matching

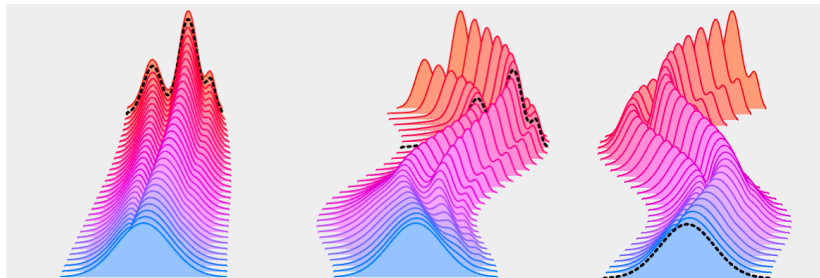
$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{f}(\mathbf{x}, t) - \mathbf{f}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

- ▶ Approximate the true vector field $\mathbf{f}(\mathbf{x}, t)$ using $\mathbf{f}_\theta(\mathbf{x}, t)$.
- ▶ Use $\mathbf{f}_\theta(\mathbf{x}, t)$ for deterministic sampling from the ODE.

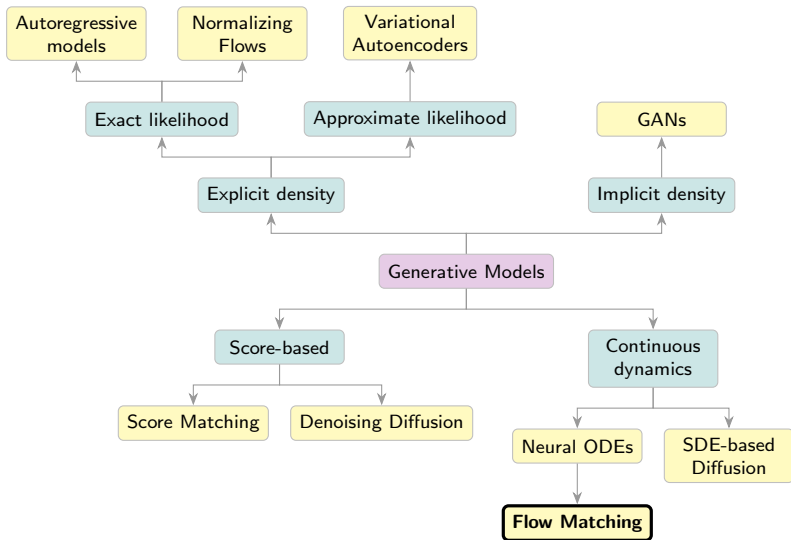
Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{f}(\mathbf{x}, t) - \mathbf{f}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

- ▶ There are infinitely many possible $\mathbf{f}(\mathbf{x}, t)$ between $p_{\text{data}}(\mathbf{x})$ and $p(\mathbf{x})$.
- ▶ The true vector field $\mathbf{f}(\mathbf{x}, t)$ is **unknown**.
- ▶ We need to select the "best" $\mathbf{f}(\mathbf{x}, t)$ and make the objective tractable.



Generative Models Taxonomy



Outline

1. Reverse SDE
2. Score-Based Generative Models Through SDEs
3. Flow Matching
4. Conditional Flow Matching

Flow Matching

Latent Variable Model

Let's introduce the latent variable $\mathbf{z}$:

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here, $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**.

Flow Matching

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$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here, $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**. The conditional probability path $p_t(\mathbf{x}|\mathbf{z})$ satisfies the KFP theorem:

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})),$$

where $\mathbf{f}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**:

Flow Matching

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where $\mathbf{f}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, t) \quad \Rightarrow \quad \frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, \mathbf{z}, t)$$

What's the relationship between $\mathbf{f}(\mathbf{x}, t)$ and $\mathbf{f}(\mathbf{x}, \mathbf{z}, t)$?

Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

Flow Matching

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})),$$

Theorem

The following vector field generates the probability path $p_t(\mathbf{x})$:

$$\mathbf{f}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})} \mathbf{f}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{f}(\mathbf{x}, \mathbf{z}, t) \frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})} d\mathbf{z}$$

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Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \frac{\partial}{\partial t} \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Flow Matching

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Proof

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Flow Matching

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Flow Matching

Flow Matching (FM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{f}(\mathbf{x}, t) - \mathbf{f}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{f}(\mathbf{x}, \mathbf{z}, t) - \mathbf{f}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

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Theorem

If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimal value of the CFM objective.

Flow Matching

Flow Matching (FM)

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If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimal value of the CFM objective.

Proof

This can be proved in a similar way as in the denoising score matching theorem.

Summary

- ▶ SDEs can be reversed using the score function.
- ▶ Score matching (NCSN) and diffusion models (DDPM) are discretizations of SDEs (variance exploding and variance preserving).
- ▶ It's possible to train continuous-in-time score-based generative models using forward and reverse SDEs.
- ▶ Discretizing the reverse SDE yields ancestral sampling of the DDPM.
- ▶ Flow matching suggests fitting the vector field directly.
- ▶ Conditional flow matching introduces the latent variable $\mathbf{z}$, reformulating the initial task in terms of conditional dynamics.